

Adaptive Forecast-Optimal Trend Estimation under Changing Time-Series Regimes

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Abstract

We study forecast-optimal trend estimation as an adaptive forecasting method, where smoothness, finite-memory window length, and finite-difference order may depend on forecast horizon and local time-series regime. The central object is therefore the full forecasting-method configuration $\Theta_{T,h}^* = (d_{T,h}^*, L_{T,h}^*, S_{T,h}^*)$, not a single penalty parameter. For the pure quadratic penalized trend, analytic first- and second-order derivatives of future forecast loss with respect to the penalty parameter permit derivative-aware numerical selection in log-penalty space. The empirical design is strictly chronological. Controlled simulations first separate forecast optimality from historical trend-recovery optimality, then study mechanisms such as persistence and horizon, and finally examine within-series regime transitions and adaptive-versus-fixed out-of-sample performance. External evidence is planned from macroeconomic series through market indices, equities, and cryptocurrency. This manuscript remains a research draft; novelty and performance claims are deferred until the literature audit and planned experiments are complete.

1 Problem formulation

Let $y \in \mathbb{R}^n$, let D_d be the order- d finite-difference matrix, and let $Q = D_d^\top D_d$. This finite-difference penalized-least-squares construction is closely related to classical graduation/smoothing methods and to the controlled-smoothness framework of Guerrero [2007], Whittaker [1923]. The pure penalized trend used for the initial analytic development is

$$\hat{t}_{\lambda,d} = \arg \min_{t \in \mathbb{R}^n} \left\{ \|y - t\|_2^2 + \lambda \|D_d t\|_2^2 \right\} = (I + \lambda Q)^{-1} y. \quad (1)$$

Writing $S_\lambda = (I + \lambda Q)^{-1}$,

$$S'_\lambda = -S_\lambda Q S_\lambda. \quad (2)$$

2 Forecast-optimal smoothness

At forecast origin T , only observations available through T may enter the fit. Let H denote the linear continuation operator from the fitted trend to the next h observations. Define

$$r_T(\lambda) = y_{T+1:T+h} - H S_\lambda y_{\text{past}}, \quad (3)$$

and

$$f_T(\lambda) = \frac{1}{h} r_T(\lambda)^\top r_T(\lambda). \quad (4)$$

The first derivative is

$$f'_T(\lambda) = \frac{2}{h} r_T(\lambda)^\top H S_\lambda Q S_\lambda y_{\text{past}}, \quad (5)$$

and the second derivative is

$$f''_T(\lambda) = \frac{2}{h} \left[\|H S_\lambda Q S_\lambda y_{\text{past}}\|_2^2 - 2 r_T(\lambda)^\top H S_\lambda Q S_\lambda Q S_\lambda y_{\text{past}} \right]. \quad (6)$$

For repeated chronological forecast origins $T_1, \dots, T_M$, the active validation objective pools future-block squared errors:

$$CV_{d,L,h}(\lambda) = \frac{\sum_{j=1}^M h_j f_{T_j}(\lambda)}{\sum_{j=1}^M h_j}. \quad (7)$$

The first and second derivatives are obtained by the same weighted aggregation of equations (5) and (6).

Because the normalized smoothness index depends on sample length, the principal inner-validation design uses a fixed-width window L for each candidate configuration. Thus all inner fits for a given (d, L) use the same sample size, and a common λ corresponds to one common normalized smoothness level. When several values of L are compared, they are scored on the same set of inner forecast origins so that changing L does not also change the validation targets.

2.1 Forecasting versus reconstruction

Randomly omitting interior observations can be useful when the scientific target is smoothing, interpolation, or trend reconstruction. It is not the validation design used for the forecasting claims in this paper, because observations later than an interior omission are then available to the smoother. Here, at every origin T , model fitting and hyperparameter selection use only information available through T ; the future block is revealed only after the forecast has been formed.

3 Numerical selection

We work with

$$\theta = \log \lambda, \quad g(\theta) = CV(e^\theta). \quad (8)$$

Then

$$g'(\theta) = \lambda CV'(\lambda), \quad g''(\theta) = \lambda CV'(\lambda) + \lambda^2 CV''(\lambda). \quad (9)$$

The planned robust procedure uses a coarse log-penalty scan only to bracket sign changes of g' , applies a bracketed root solver to every detected interval using a bracketed zero-finding method in the spirit of Brent [1971], classifies the stationary points, and compares all local minima with the domain boundaries. Dense grid search and Newton iterations are retained as diagnostic and computational benchmarks.

4 Research questions

At forecast origin T and horizon h , define the forecast-method configuration

$$\Theta_{T,h}^* = (d_{T,h}^*, L_{T,h}^*, S_{T,h}^*) = G(h, X_T, \mathcal{C}), \quad (10)$$

where d is difference order, L is finite-memory estimation-window length, S is normalized smoothness, X_T summarizes the local state, and $\mathcal{C}$ denotes the series class. Candidate components of X_T include volatility, persistence, trend strength, local roughness, and break evidence.

The paper has three empirical questions. First, does the forecast-optimal configuration differ from the configuration that best reconstructs a known historical latent trend? Second, which mechanisms make $\Theta_{T,h}^*$ vary with horizon and local state? Third, does adapting the forecasting method to those states improve untouched future forecasts relative to strong fixed configurations?

The persistence/horizon experiment is one mechanism study addressing the second question. Its oracle AR-residual-aware forecast is diagnostic only: it must not be interpreted as the definition of the paper or as the proposed real-data model. Local regime is deliberately broader than persistence alone.

5 Experimental design

The planned sequence is:

1. controlled simulations separating forecast and recovery targets;
2. mechanism studies for persistence, horizon, signal-to-noise structure, roughness, endpoints, and related effects;
3. within-series regime-transition experiments tracking the joint path of (d^*, L^*, S^*) and adaptation delay;
4. adaptive-versus-fixed untouched out-of-sample comparisons;
5. lower-frequency macroeconomic series;
6. broad market indices and ETFs;
7. individual equities;
8. cryptocurrency.

All real-data experiments use nested chronological evaluation. At an outer origin, only the available history is passed to the inner selector; after (d, L, λ) is selected, the chosen model is refitted on the selected window and forecasts the untouched future block. For financial level forecasts, the mandatory baseline is

$$\hat{y}_{T+k|T}^{(0)} = y_T, \quad k = 1, \dots, h, \quad (11)$$

and we report the relative pooled forecast error

$$RMSFE_{\text{rel}} = \frac{RMSFE_{\text{method}}}{RMSFE_{\text{no-change}}}. \quad (12)$$

Return or trend-change targets will be considered separately.

6 Scope

The current paper does not study portfolio utility or decision-optimal smoothing. Those questions remain possible future work built on the same `trend_estimation` library.

References

- Richard P. Brent. An algorithm with guaranteed convergence for finding a zero of a function. *The Computer Journal*, 14(4):422–425, 1971. doi: 10.1093/comjnl/14.4.422. URL <https://doi.org/10.1093/comjnl/14.4.422>.
- Victor M. Guerrero. Time series smoothing by penalized least squares. *Statistics & Probability Letters*, 77(12):1225–1234, 2007. doi: 10.1016/j.spl.2007.03.006. URL <https://doi.org/10.1016/j.spl.2007.03.006>.
- Edmund T. Whittaker. On a new method of graduation. *Proceedings of the Edinburgh Mathematical Society*, 41:63–75, 1923. doi: 10.1017/S0013091500077853. URL <https://doi.org/10.1017/S0013091500077853>.